Information Gathering Using OSINT Tools

WHOIS • theHarvester • nslookup • Google Dorking • Subfinder

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1. Introduction

Information gathering is an early phase of a cybersecurity assessment. It involves collecting and reviewing publicly available information about a domain, organization, or its internet-facing presence. Open-Source Intelligence (OSINT) refers to information collected from public sources and analyzed for a specific purpose.

This project demonstrates five beginner-friendly tools and techniques: WHOIS, theHarvester, nslookup, Google Dorking, and Subfinder. The practical work focuses on passive collection and documentation rather than exploitation or unauthorized access.

Project Details

Item	Description
Project Title	Information Gathering in Cybersecurity Using OSINT Tools
Platform	Kali Linux
Category	Cybersecurity / Open-Source Intelligence (OSINT)
Level	Beginner
Target	A domain owned by the student or explicitly authorized for assessment
Main Output	Command outputs, observations, screenshots, and recommendations

2. Project Objectives and Scope

- Understand the role of OSINT and passive reconnaissance in cybersecurity.
- Collect public domain registration information with WHOIS.
- Use theHarvester to review publicly discoverable hostnames and other available information.
- Query DNS records with nslookup.
- Use Google search operators to locate publicly indexed pages and documents.
- Discover subdomains with Subfinder using passive sources.
- Document findings clearly and identify the limitations of each tool.

Scope and Authorization

Use a domain you own or have explicit permission to assess. example.com is a reserved example domain that can be used to understand commands, but its results may be limited and will not represent a full real-world assessment. Do not attempt to log in to discovered services, exploit systems, bypass access controls, or access private information.

3. Lab Setup and Tool Installation

Open a terminal in Kali Linux and update the package lists:

```
sudo apt update
sudo apt upgrade
```

Install the initial tools:

```
sudo apt install whois dnsutils theharvester
```

Check tool help or availability. Some versions may not support the version flag; use -h when needed.

```
whois --version
theHarvester -h
nslookup -help
```

Install Subfinder

```
sudo apt install subfinder
subfinder -version
```

If Kali cannot find the package or the packaged version is unsuitable, follow the official ProjectDiscovery installation guide: https://docs.projectdiscovery.io/opensource/subfinder/install

Prepare an Evidence Folder

```
mkdir -p ~/information-gathering-project/evidence
cd ~/information-gathering-project
```

Replace example.com in the commands with your authorized assessment domain.

4. Practical — WHOIS Enumeration

A. Description

WHOIS is a protocol and lookup service used to retrieve domain registration information. Depending on the registry and privacy settings, output may include the registrar, registration dates, name servers, status, and registrant details.

B. Commands to Perform

```
whois example.com
whois example.com | tee evidence/whois.txt
```

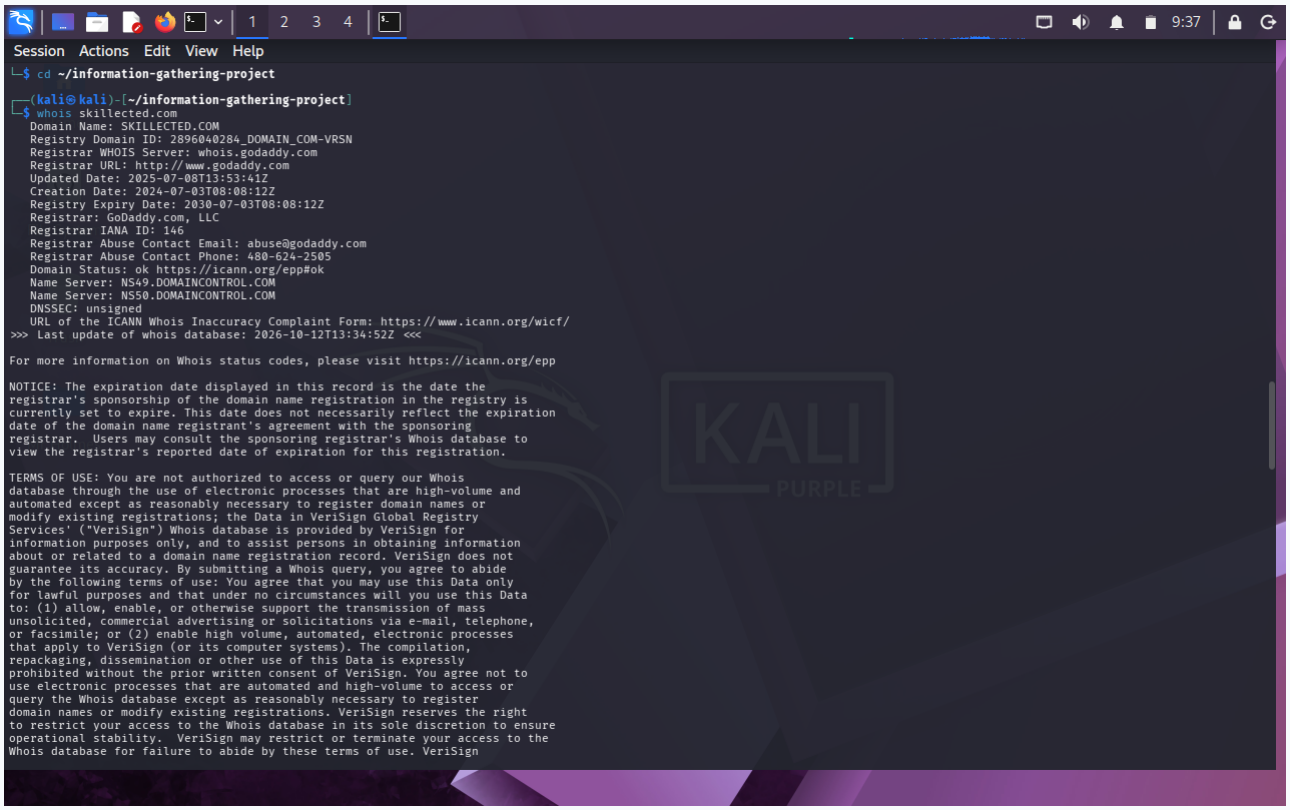
C. Observed Information

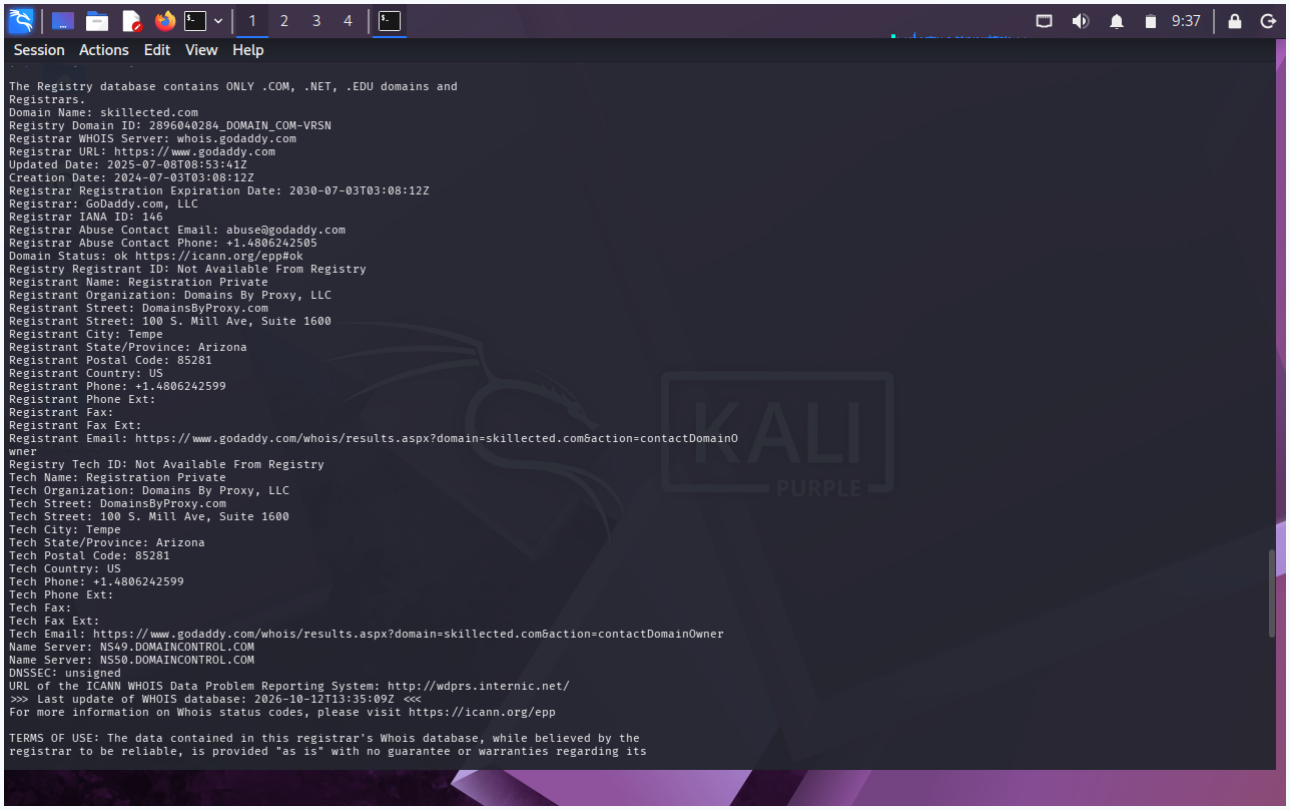
Field / Item	What to Record
Registrar	Company managing the domain registration, if shown
Creation Date	Date the domain was registered, if provided
Expiry Date	Registration expiry date, if provided
Name Server	DNS servers associated with the domain
Domain Status	Registration status values
Registrant	Owner information only if publicly available

D. Purpose

- Understand domain registration details.
- Identify the registrar and configured name servers.
- Collect background information for an authorized assessment.
- Record which information is hidden or unavailable.

E. Output Showing





5. Practical — theHarvester

A. Description

theHarvester is an OSINT tool that gathers publicly available information about a domain using supported search engines and other public sources. Results depend on the selected source, network access, configuration, and source limits.

B. Commands to Perform

```
theHarvester -h
theHarvester -d example.com -b crtsh
theHarvester -d example.com -b crtsh >
evidence/theharvester.txt 2>&1
```

The crt.sh source relates to public certificate-transparency data. If the selected source is unavailable in your installed version, consult the tool's help output and use a supported source.

C. Observed Information

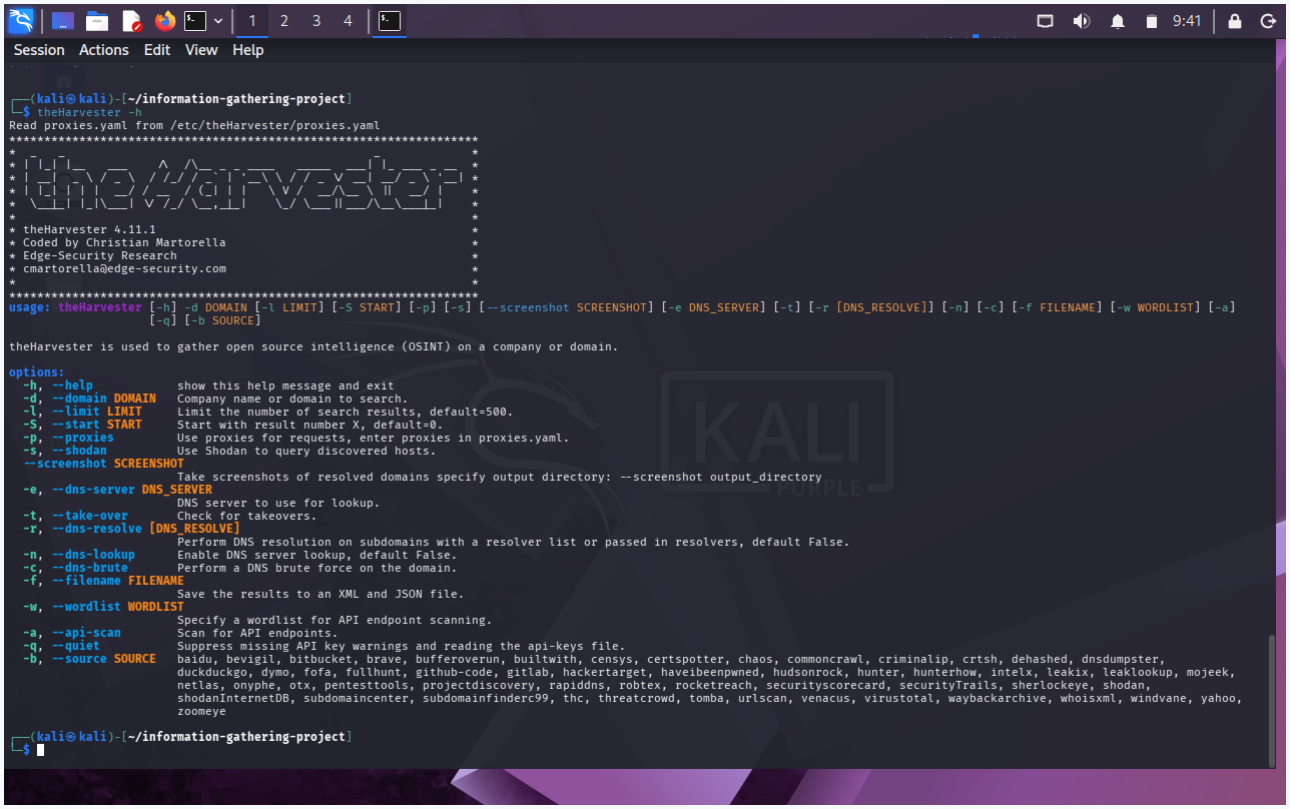
Field / Item	What to Record
Emails	Publicly discovered email addresses, if any
Hosts	Hostnames found by supported sources
IP Addresses	Addresses associated with discovered hosts, if returned
Sources	Data sources queried by the tool
Errors / Limits	Unavailable sources, rate limits, or lookup errors

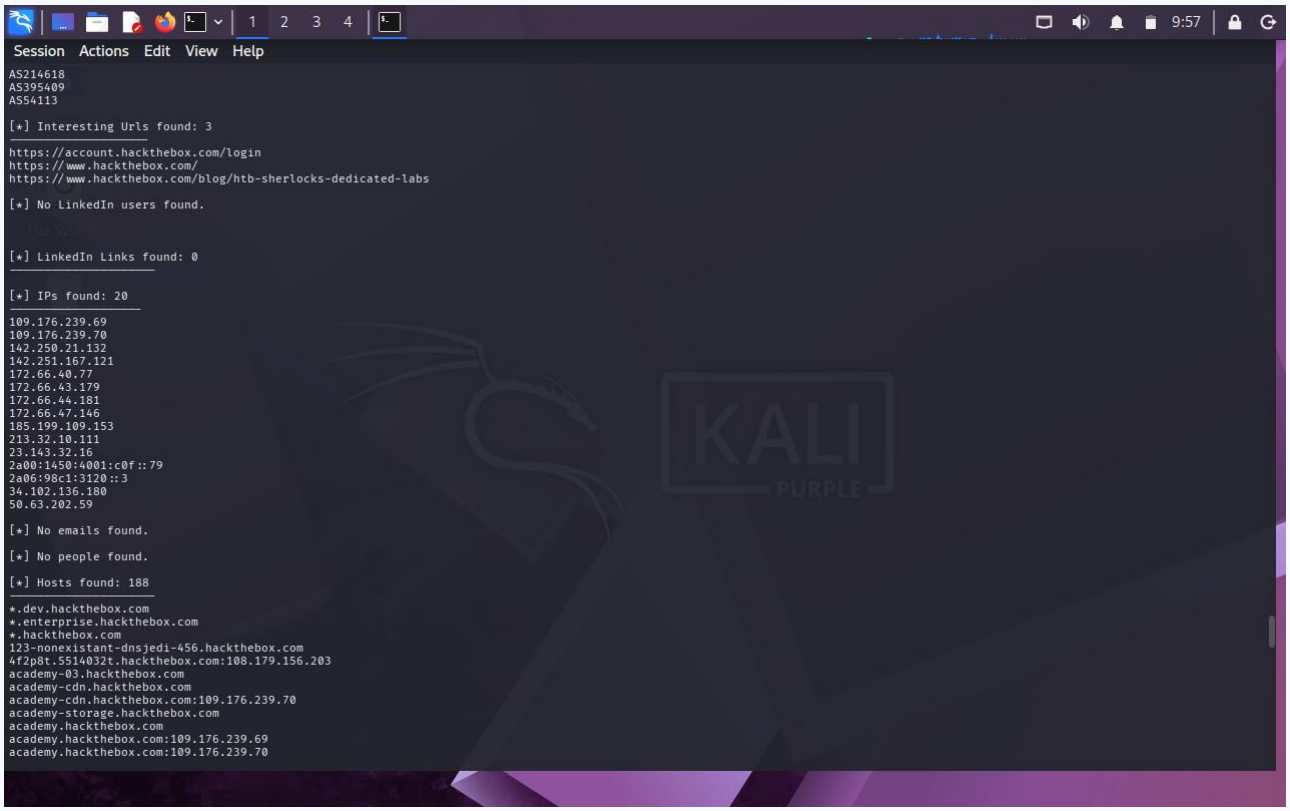
D. Purpose

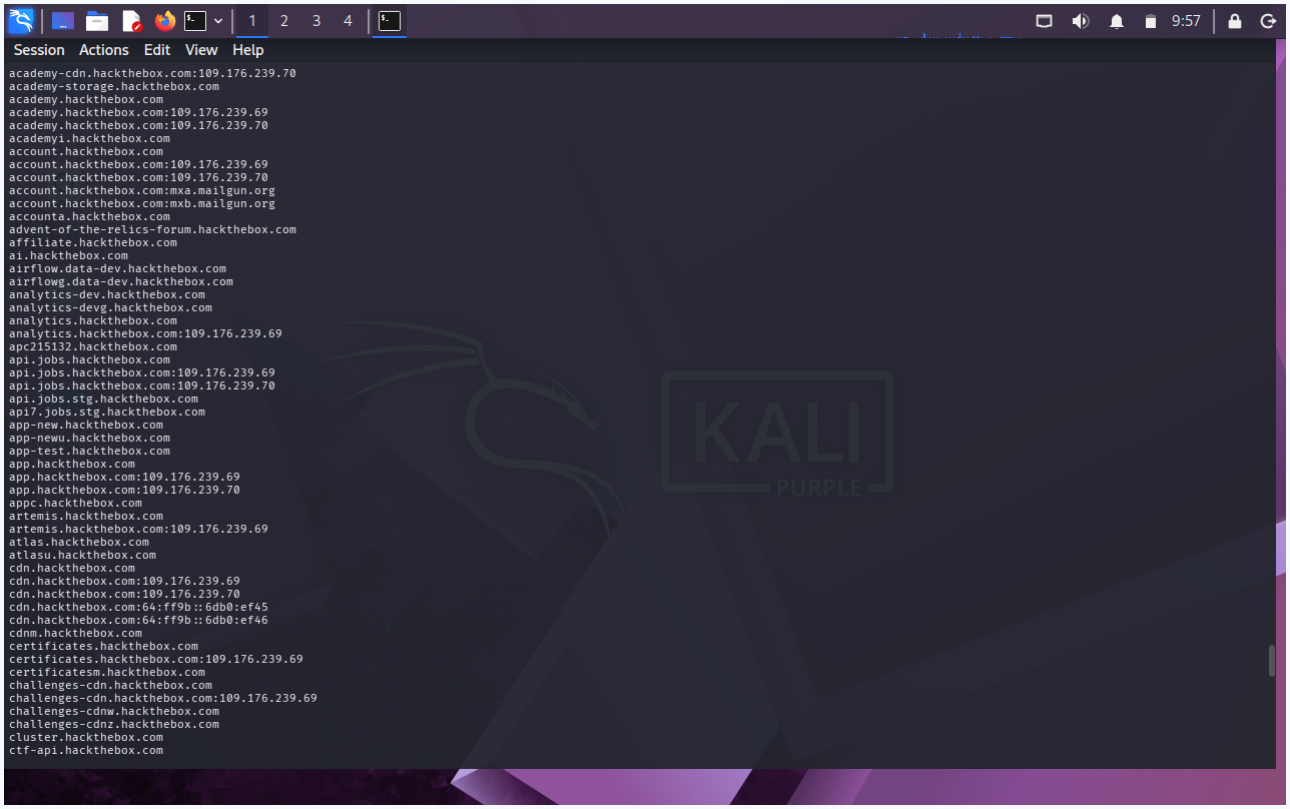
- Learn how OSINT tools collect public information.
- Review the public digital footprint of an authorized domain.
- Understand that results can be incomplete and source-dependent.

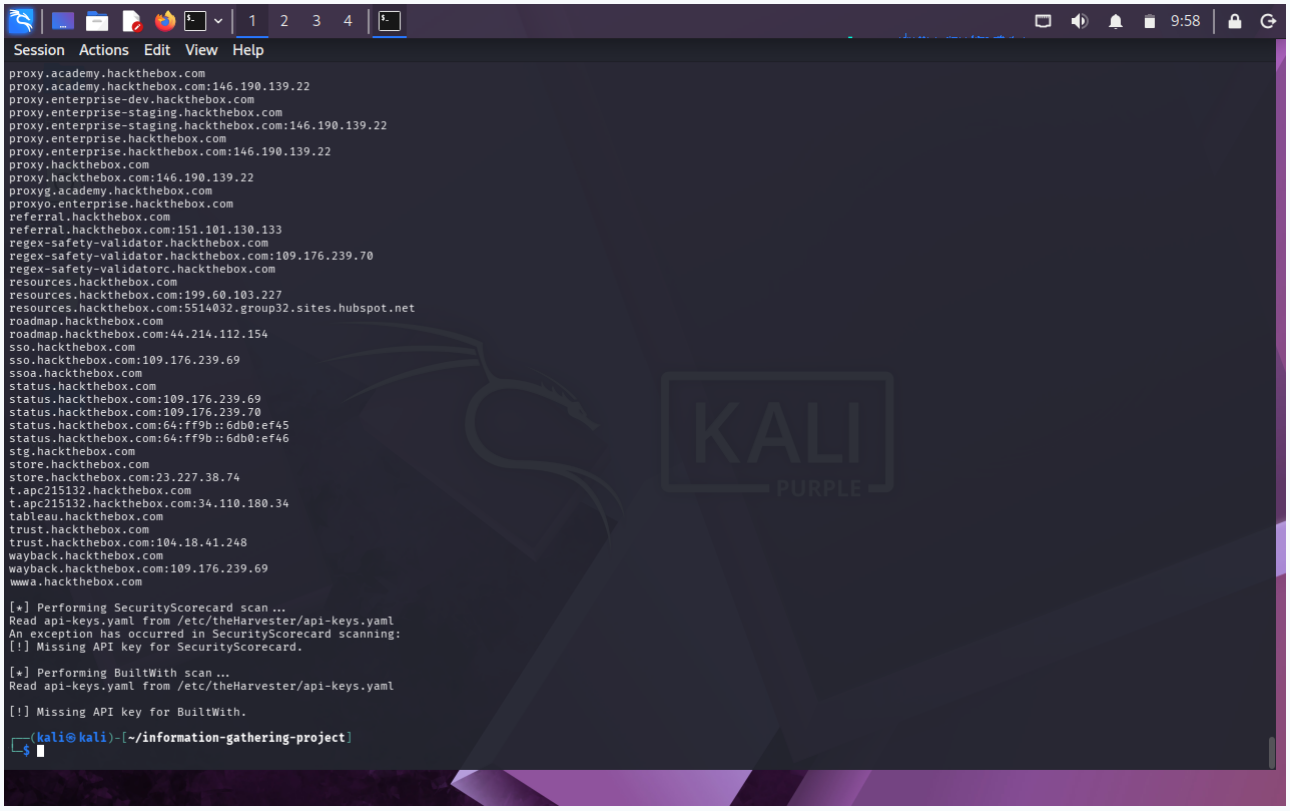
- Record useful findings without publishing sensitive details.

E. Output Showing






1 2 3 4
9:57
Session Actions Edit View Help
academy-cdn.hackthebox.com:109.176.239.70
academy-storage.hackthebox.com
academy.hackthebox.com
academy.hackthebox.com:109.176.239.69
academy.hackthebox.com:109.176.239.70
academyi.hackthebox.com
account.hackthebox.com
account.hackthebox.com:109.176.239.69
account.hackthebox.com:109.176.239.70
account.hackthebox.com:mxa.mailgun.org
account.hackthebox.com:mxb.mailgun.org
accounta.hackthebox.com
advent-of-the-relics-forum.hackthebox.com
affiliate.hackthebox.com
ai.hackthebox.com
airflow.data-dev.hackthebox.com
airflowg.data-dev.hackthebox.com
analytics-dev.hackthebox.com
analytics-devg.hackthebox.com
analytics.hackthebox.com
analytics.hackthebox.com:109.176.239.69
apc215132.hackthebox.com
api.jobs.hackthebox.com
api.jobs.hackthebox.com:109.176.239.69
api.jobs.hackthebox.com:109.176.239.70
api.jobs.stg.hackthebox.com
api7.jobs.stg.hackthebox.com
app-new.hackthebox.com
app-newu.hackthebox.com
app-test.hackthebox.com
app.hackthebox.com
app.hackthebox.com:109.176.239.69
app.hackthebox.com:109.176.239.70
appc.hackthebox.com
artemis.hackthebox.com
artemis.hackthebox.com:109.176.239.69
atlas.hackthebox.com
atlasu.hackthebox.com
cdn.hackthebox.com
cdn.hackthebox.com:109.176.239.69
cdn.hackthebox.com:109.176.239.70
cdn.hackthebox.com:64:ff9b::6db0:ef45
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cdnm.hackthebox.com
certificates.hackthebox.com
certificates.hackthebox.com:109.176.239.69
certificatesm.hackthebox.com
challenges-cdn.hackthebox.com
challenges-cdn.hackthebox.com:109.176.239.69
challenges-cdnw.hackthebox.com
challenges-cdnz.hackthebox.com
cluster.hackthebox.com
ctf-api.hackthebox.com


1 2 3 4
9:58
Session Actions Edit View Help
proxy.academy.hackthebox.com
proxy.academy.hackthebox.com:146.190.139.22
proxy.enterprise-dev.hackthebox.com
proxy.enterprise-staging.hackthebox.com
proxy.enterprise-staging.hackthebox.com:146.190.139.22
proxy.enterprise.hackthebox.com
proxy.enterprise.hackthebox.com:146.190.139.22
proxy.hackthebox.com
proxy.hackthebox.com:146.190.139.22
proxyg.academy.hackthebox.com
proxyo.enterprise.hackthebox.com
referral.hackthebox.com
referral.hackthebox.com:151.101.130.133
regex-safety-validator.hackthebox.com
regex-safety-validator.hackthebox.com:109.176.239.70
regex-safety-validatorc.hackthebox.com
resources.hackthebox.com
resources.hackthebox.com:199.60.103.227
resources.hackthebox.com:5514032.group32.sites.hubspot.net
roadmap.hackthebox.com
roadmap.hackthebox.com:44.214.112.154
sso.hackthebox.com
sso.hackthebox.com:109.176.239.69
ssoa.hackthebox.com
status.hackthebox.com
status.hackthebox.com:109.176.239.69
status.hackthebox.com:109.176.239.70
status.hackthebox.com:64:ff9b::6db0:ef45
status.hackthebox.com:64:ff9b::6db0:ef46
stg.hackthebox.com
store.hackthebox.com
store.hackthebox.com:23.227.38.74
t.apc215132.hackthebox.com
t.apc215132.hackthebox.com:34.110.180.34
tableau.hackthebox.com
trust.hackthebox.com
trust.hackthebox.com:104.18.41.248
wayback.hackthebox.com
wayback.hackthebox.com:109.176.239.69
wwwa.hackthebox.com
[*] Performing SecurityScorecard scan ...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
An exception has occurred in SecurityScorecard scanning:
[!] Missing API key for SecurityScorecard.
[*] Performing BuiltWith scan ...
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for BuiltWith.
(kali㉿kali)-[~/information-gathering-project]
$

6. Practical — nslookup

A. Description

nslookup is a DNS lookup utility. It queries the Domain Name System (DNS) to resolve domain names and retrieve DNS records.

B. Commands to Perform

```
nslookup example.com
nslookup -type=MX example.com
nslookup -type=NS example.com
nslookup -type=TXT example.com
nslookup example.com > evidence/nslookup.txt
```

Some records may not exist, and DNS responses can vary by resolver. An absent record is not automatically a security problem.

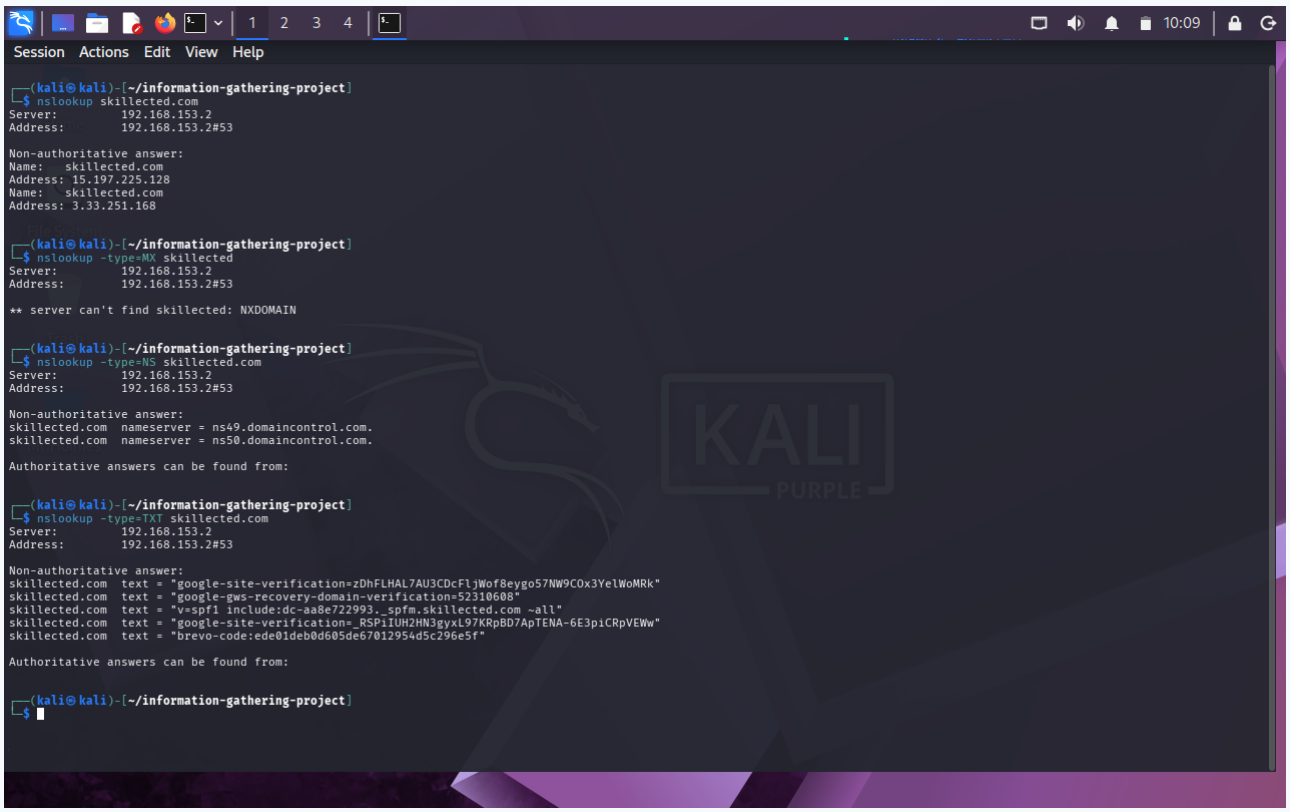
C. Observed Information

Field / Item	What to Record
A	IPv4 address
AAAA	IPv6 address
MX	Mail server records
NS	Authoritative name server records
TXT	Text records, which may include SPF or domain verification data

D. Purpose

- Understand domain-to-IP resolution.
- Identify publicly configured mail and name servers.
- Examine DNS configuration for an authorized domain.
- Learn how DNS supports web and email services.

E. Output Showing



7. Practical — Google Dorking

A. Description

Google Dorking uses advanced search operators to locate publicly indexed information. It is a search technique rather than a separate software tool. Search results vary over time and may differ by region or account settings.

B. Search Queries to Perform

Purpose	Example Query
Find indexed pages	site:example.com
Find public PDF documents	site:example.com filetype:pdf
Find public policy pages	site:example.com (privacy OR security) policy
Find public documentation	site:example.com documentation

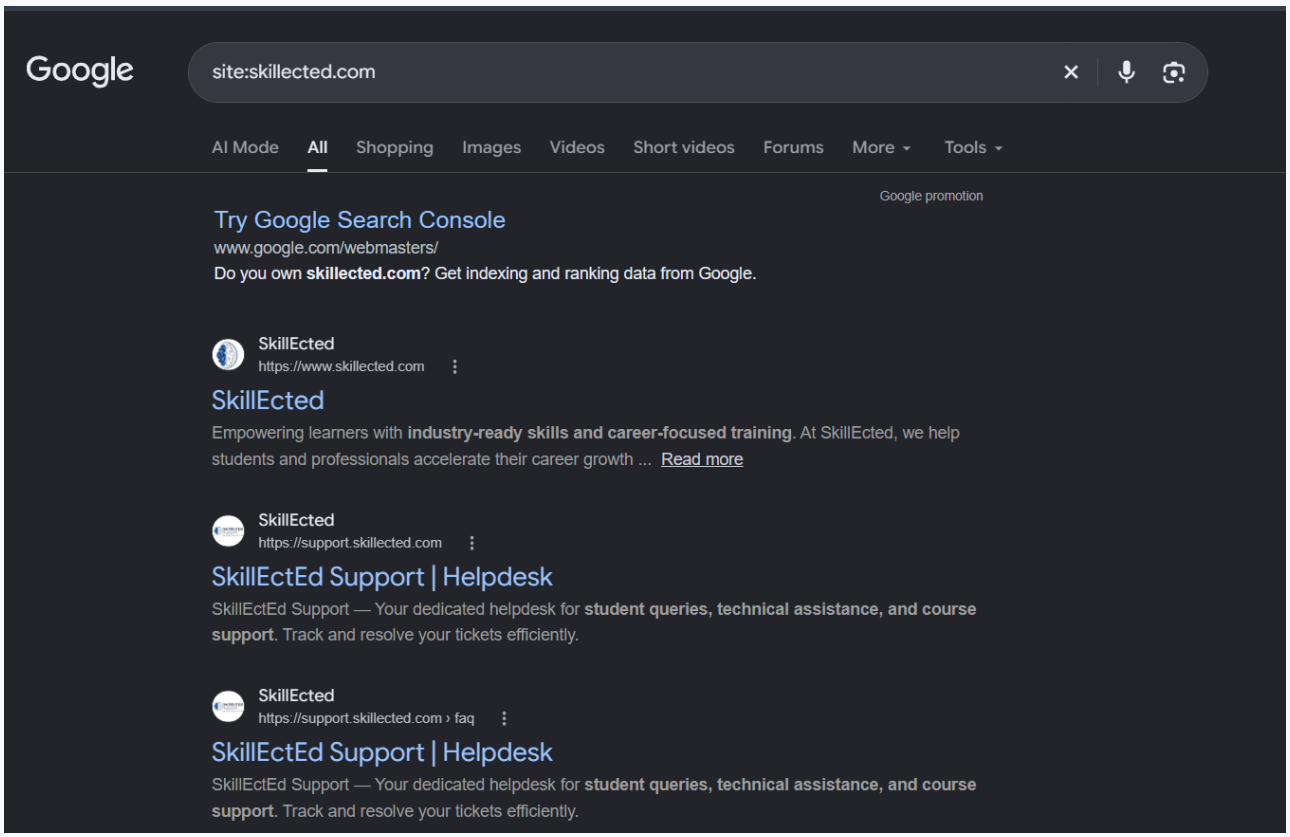
C. Observed Information

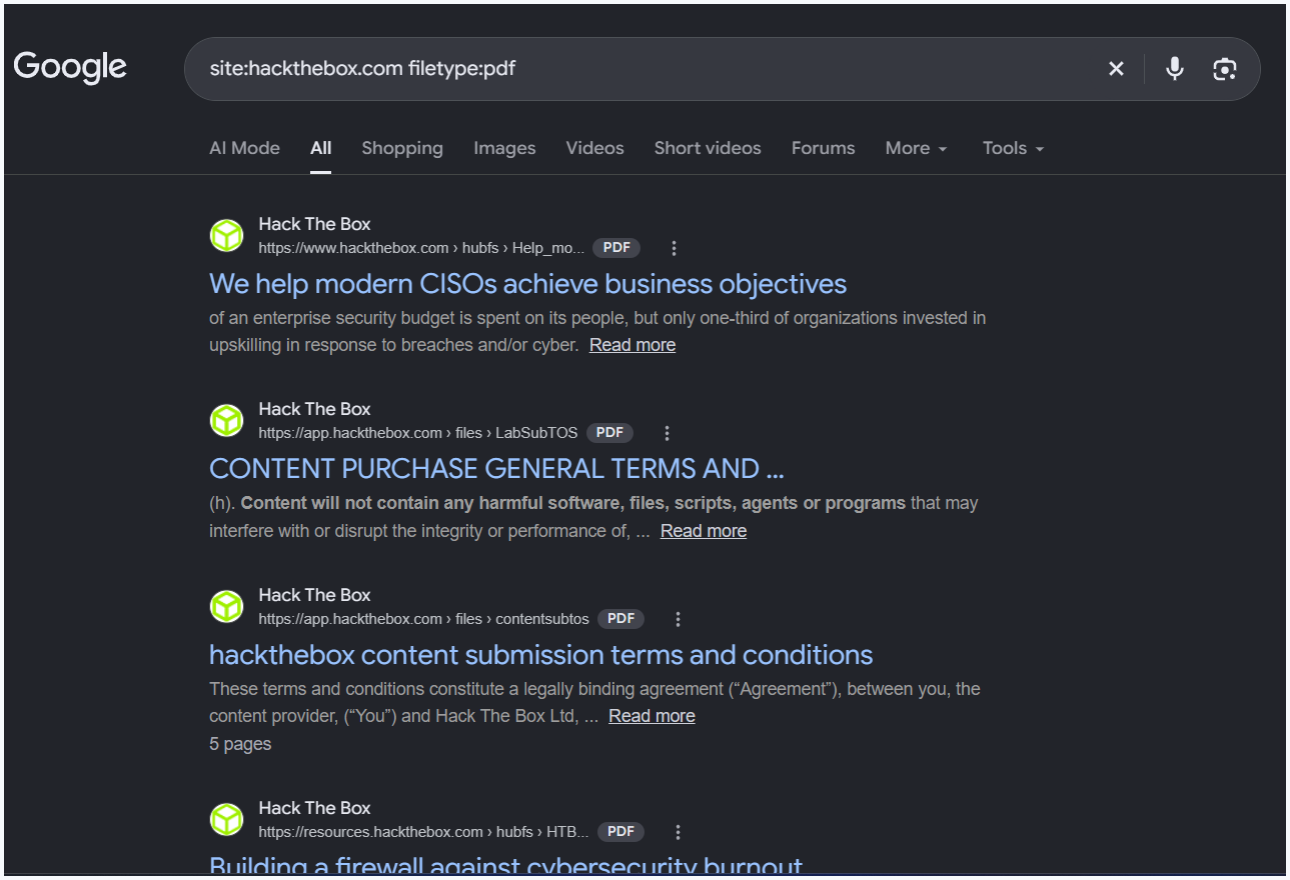
- Publicly indexed web pages.
- Public PDF files and documentation.
- Public policy pages and descriptions of services.
- Potentially outdated public content that the domain owner may wish to review.

D. Purpose

- Learn advanced search techniques.
- Understand how search engines index publicly accessible content.
- Identify unnecessary public information exposure.
- Recommend improvements to public information management.

E. Output Showing





8. Practical — Subfinder

A. Description

Subfinder is a subdomain discovery tool from ProjectDiscovery. It primarily uses passive sources to identify subdomains associated with a domain. A discovered name does not prove that the host is currently active or vulnerable.

B. Commands to Perform

```
subfinder -version
subfinder -d example.com
subfinder -d example.com -silent
subfinder -d example.com -silent -o
evidence/subdomains.txt
wc -l evidence/subdomains.txt
```

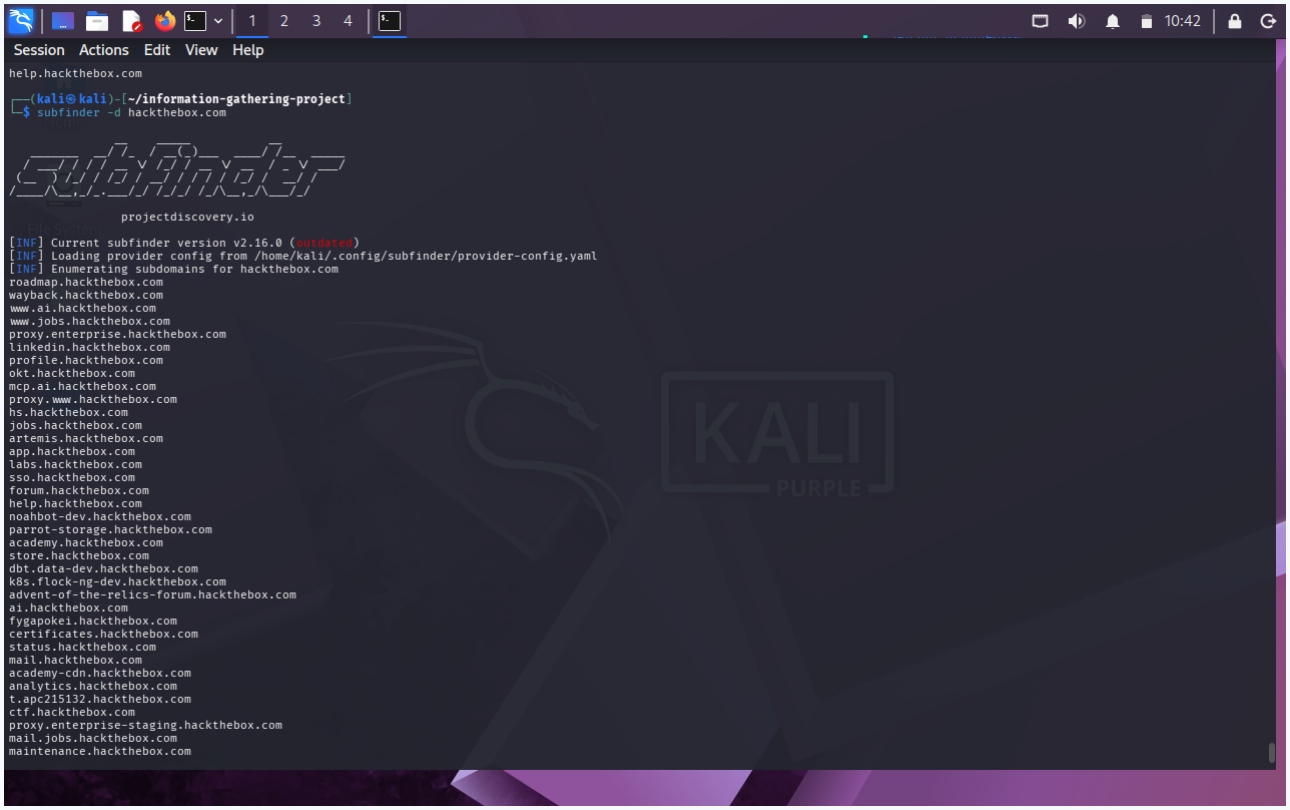
C. Observed Information

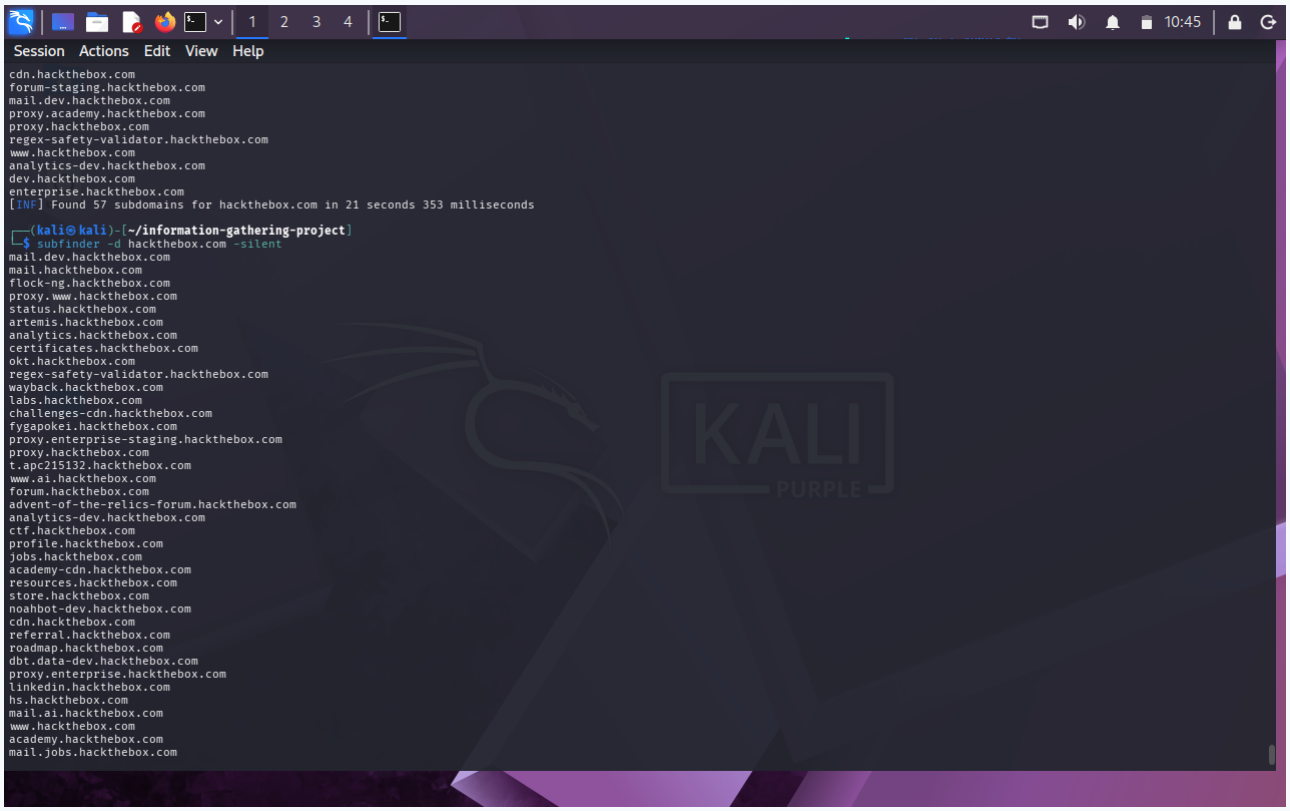
Field / Item	What to Record
Subdomains	Names associated with the target domain
Number of Results	Count of discovered names
Source Coverage	Results depend on sources and configuration
Limitations	No results do not prove that no subdomains exist

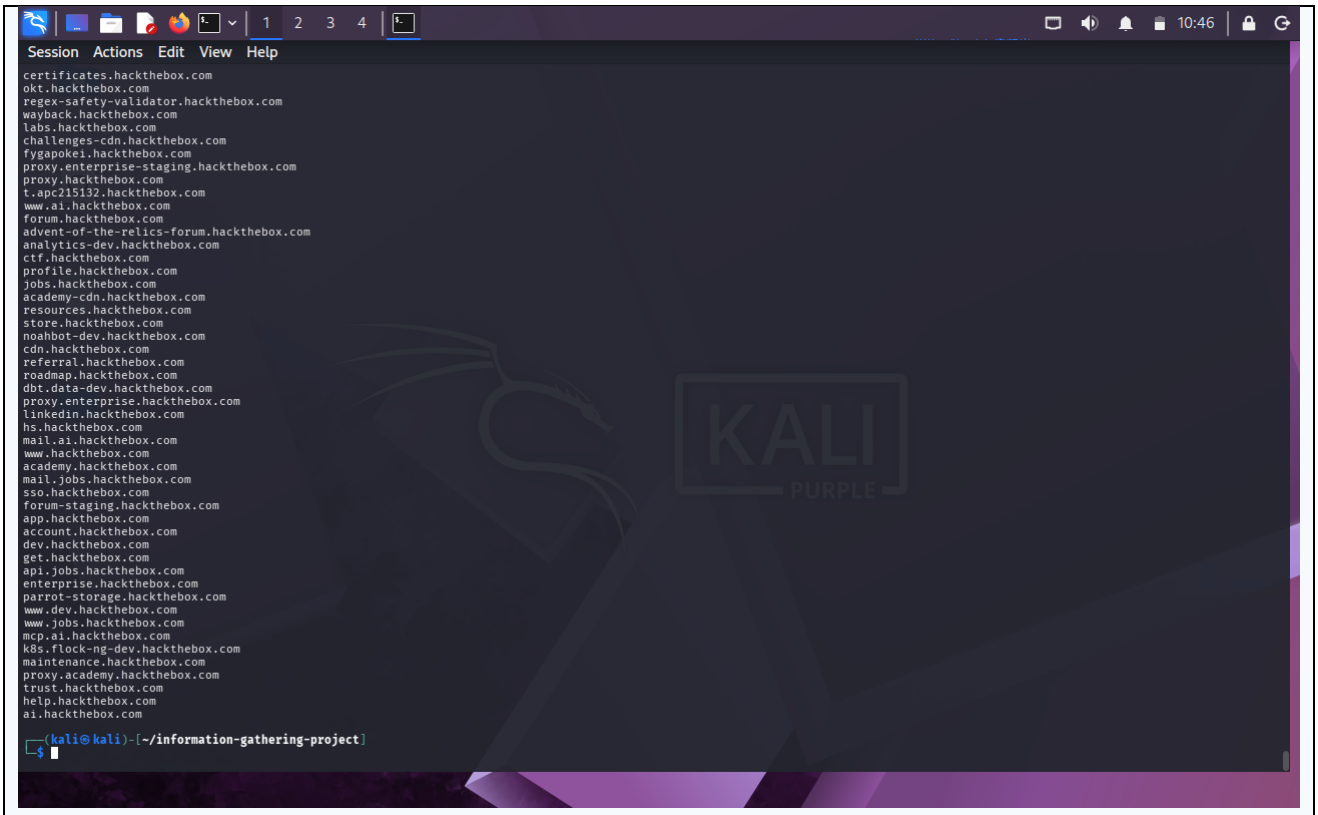
D. Purpose

- Discover publicly known subdomains.
- Understand the size of an organization's public namespace.
- Help build an inventory for an authorized assessment.
- Identify unexpected names for the domain owner to review.

E. Output Showing






1 2 3 4
10:46
Session Actions Edit View Help
certificates.hackthebox.com
okt.hackthebox.com
regex-safety-validator.hackthebox.com
wayback.hackthebox.com
labs.hackthebox.com
challenges-cdn.hackthebox.com
fygapokei.hackthebox.com
proxy.enterprise-staging.hackthebox.com
proxy.hackthebox.com
t.apc215132.hackthebox.com
www.ai.hackthebox.com
forum.hackthebox.com
advent-of-the-relics-forum.hackthebox.com
analytics-dev.hackthebox.com
ctf.hackthebox.com
profile.hackthebox.com
jobs.hackthebox.com
academy-cdn.hackthebox.com
resources.hackthebox.com
store.hackthebox.com
noahbot-dev.hackthebox.com
cdn.hackthebox.com
referral.hackthebox.com
roadmap.hackthebox.com
dbt.data-dev.hackthebox.com
proxy.enterprise.hackthebox.com
linkedin.hackthebox.com
hs.hackthebox.com
mail.ai.hackthebox.com
www.hackthebox.com
academy.hackthebox.com
mail.jobs.hackthebox.com
sso.hackthebox.com
forum-staging.hackthebox.com
app.hackthebox.com
account.hackthebox.com
dev.hackthebox.com
get.hackthebox.com
api.jobs.hackthebox.com
enterprise.hackthebox.com
parrot-storage.hackthebox.com
www.dev.hackthebox.com
www.jobs.hackthebox.com
mcp.ai.hackthebox.com
k8s.flock-ng-dev.hackthebox.com
maintenance.hackthebox.com
proxy.academy.hackthebox.com
trust.hackthebox.com
help.hackthebox.com
ai.hackthebox.com
(kali㉿kali)-[~/information-gathering-project]
$
KALI
PURPLE

9. Evidence Management and Observation Table

Tool	What to Record
WHOIS	Registrar, dates, name servers, status
theHarvester	Hosts, emails if returned, source errors
nslookup	Address and DNS record results
Google Dorking	Query and relevant public results
Subfinder	Discovered names and count

10. Methodology

1. Define the scope and confirm permission to assess the selected domain.
2. Collect public registration information using WHOIS.
3. Use theHarvester to query supported public sources.
4. Query address and DNS records with nslookup.
5. Use safe Google search operators to review publicly indexed pages and documents.
6. Use Subfinder for passive subdomain discovery.
7. Save outputs, capture screenshots, and record observations.
8. Review the findings, note limitations, and prepare recommendations for the domain owner.

11. Security, Ethics, and Limitations

- Authorization: Assess only domains you own or have explicit permission to examine.
- Passive does not mean risk-free: requests to third-party sources may be logged, and some tools may contact data providers.
- No exploitation: This project is limited to information gathering; do not attempt to access accounts, exploit services, or bypass controls.
- Protect evidence: Do not publish personal data, credentials, confidential details, or sensitive screenshots.
- Interpret carefully: Results may be outdated, incomplete, rate-limited, or affected by DNS and search-engine behavior.
- Respect laws, site terms, and organizational policies.

12. Conclusion

This project demonstrates five information-gathering techniques in cybersecurity: WHOIS Enumeration, theHarvester, nslookup, Google Dorking, and Subfinder. These tools help security practitioners review public domain registration details, DNS records, indexed pages, and known subdomains.

The practical builds familiarity with Open-Source Intelligence (OSINT), passive reconnaissance, evidence collection, and clear reporting. Findings should be interpreted cautiously, documented accurately, and used only within an authorized scope.